

A Graph Analysis of Jeffrey Epstein’s Aviation Criminal Network

1 Introduction

The Jeffrey Epstein scandal represents one of the most complex cases of systemic elite corruption in recent years, underpinned by a massive organized network that bridged geopolitical power, corporate capital and logistics. Central to this operation was his fleet of private aircraft, most notably a Boeing 727 that became known in the press as the “Lolita Express”. In 2024, thousands of pages of legal documents were released against Epstein and his associates. These records include flight logs, sworn depositions, email correspondence, court filings, and investigative reports, collectively forming one of the richest forensic datasets ever made publicly available about an organized crime network.

This study draws on a graph constructed by [Epstein Document Explorer](#), which used large language models to extract named entities, relationships, and metadata from the documents released by House Oversight in an automated manner. The full graph contains 546,648 nodes and 1,432,277 edges across multiple node labels and edge types.

The study will focus on the aviation subnetwork only, the web of entities connected through flight-related interactions. The core entity-level relationship used in this study is `(:Entity)-[:RELATED_TO]->(:Entity)`. This allows analysis at the network level, examining who is connected to whom, through what types of interactions, and how frequently. The aircrafts were not merely a mode of transport, but also the physical infrastructure through which Epstein cultivated relationships, exerted influence, and facilitated abuse. By isolating the aviation subgraph, we aim to:

- a. Identify the key actors in Epstein’s flight network and their structural roles.
- b. Detect cohesive communities within the aviation subgraph and interpret what each community represents.
- c. Quantify influence and brokerage using graph centrality metrics.

2 Data Preprocessing and Exploration

2.1 Data Preprocessing

The full graph was hosted in Memgraph, a high-performance graph database supporting the Cypher query language. To extract only aviation-related subgraph, the following Cypher query was used:

```

MATCH (a:Entity)-[r:RELATED_TO]->(b:Entity)
WHERE toLower(r.action) CONTAINS 'flew'
      OR toLower(r.action) CONTAINS 'plane'
      OR toLower(r.action) CONTAINS 'flight'
      OR toLower(r.action) CONTAINS 'aviation'
      OR toLower(r.action) CONTAINS 'aircraft'
RETURN a, r, b

```

This query matches all directed `RELATED_TO` edges whose `action` property contains any aviation-related keywords. The resulting subgraph was exported and loaded in Gephi for further analysis and visualization.

The subgraph consists of 253 nodes and 246 unique directed edges, representing a focused but highly informative slice of the broader Epstein network. Each unique edge carries a weight attribute derived from the frequency of documented interactions between two entities. Edges with higher weights (e.g., Bill Clinton → Jeffrey Epstein with weight 27, or Jeffrey Epstein → Bill Clinton with weight 25) represent relationships documented repeatedly across multiple sources. In Gephi’s ForceAtlas2 layout algorithm, edge weight directly influences the gravitational attraction between nodes, with more heavily documented relationships pulling nodes closer together, while weakly documented links result in greater separation.

2.2 Data Exploration

Table 1 shows the most common actions in the aviation subgraph, which describe the nature of the relationship between two entities and were extracted directly from the source documents by LLM parsing. Variations of “flew” reflect the direct co-travel relationships, while the presence of “testified that was passenger on plane with” and “provided flight logs to” underscores that the dataset includes not only the original travel records but also subsequent legal proceedings in which these events were formally attested. During the topological aggregation in Gephi, the qualitative action column utilized a ‘First’ selection strategy which retains the text of the initial record within identical pairs. Therefore, this baseline reference table merely serves as a sample to qualitatively contextualize the broad behavioral domains extracted from the underlying source data.

Table 1: Top 10 Most Frequent Actions in the Aviation Subgraph

Rank	Action	Count
1	flew on	12
2	flew to	10
3	flew with	10
4	flew	10
5	testified that was passenger on plane with	8
6	traveled on aircraft with	7
7	flew as passenger on	4
8	provided flight logs to	4
9	traveled on flight #1073 with	4
10	flew_with	2

Table 2 shows the node pairs with the highest edge weight (i.e., the most heavily documented individual relationships). The bidirectional nature of the Clinton–Epstein connection (Clinton flying on Epstein’s jet; Epstein providing aircraft to Clinton) reveals a relationship documented from multiple angles across different documents. The high weight for Larry Vidoski → Jeffrey Epstein reflects Vidoski’s role as Epstein’s personal pilot with repeated operational communications, while Barry Cohen and Alan M. Dershowitz managed the legal defense for Epstein.

Table 2: Top 6 Heaviest Individual Edges by Weight

Rank	Source	Target	Edge Weight
1	Bill Clinton	Jeffrey Epstein	27
2	Jeffrey Epstein	Bill Clinton	25
3	Jeffrey Epstein	Kevin Spacey	11
4	Barry Cohen	Jeffrey Epstein	11
5	Larry Vidoski	Jeffrey Epstein	9
6	Alan M. Dershowitz	Jeffrey Epstein	8

Though the subgraph is centered on human actors, the Entity nodes in the graph encompass rich types. This variety reflects the LLM extraction process, which captured not only named individuals but also organizations, locations, aircraft, legal instruments, and abstract references. The inclusion of non-human entities enriches the network by providing contextual anchors.

Table 3: Node Type Categories and Example Nodes

Category	Example Nodes
People	Jeffrey Epstein, Bill Clinton, Alan M. Dershowitz, Ghislaine Maxwell, Kevin Spacey, Prince Andrew, Larry Vidoski
Locations	Teterboro Airport, St Thomas
Aircraft	G550 aircraft, Boeing B727
Organizations	Aeroflot, United States District Court
Evidence	flight logs
Groups	young girls, members of Mr. Epstein’s legal team, federal and state authorities

3 Community Mining

3.1 Algorithm

Community detection was performed in Gephi using the Louvain Modularity algorithm, which is Gephi’s built-in community detection method. The Louvain algorithm iteratively optimizes modularity, a measure of how well the network decomposes into dense internally connected groups with sparse inter-group connections. This algorithm identified multiple distinct communities in the aviation subgraph, each assigned a unique modularity class ID. The most relevant clusters are:

Table 4: Key Communities Identified by the Louvain Algorithm

Community	Modularity Class	Size (nodes)	Key Figure
Blue (Core Network)	1	83	Jeffrey Epstein
Green (Legal Cluster)	3	29	Alan M. Dershowitz
Yellow (VIP Passengers)	19	11	Larry Visoki
Purple (Pilot Operations)	13	7	Larry Visoski

3.2 Full Network Visualization

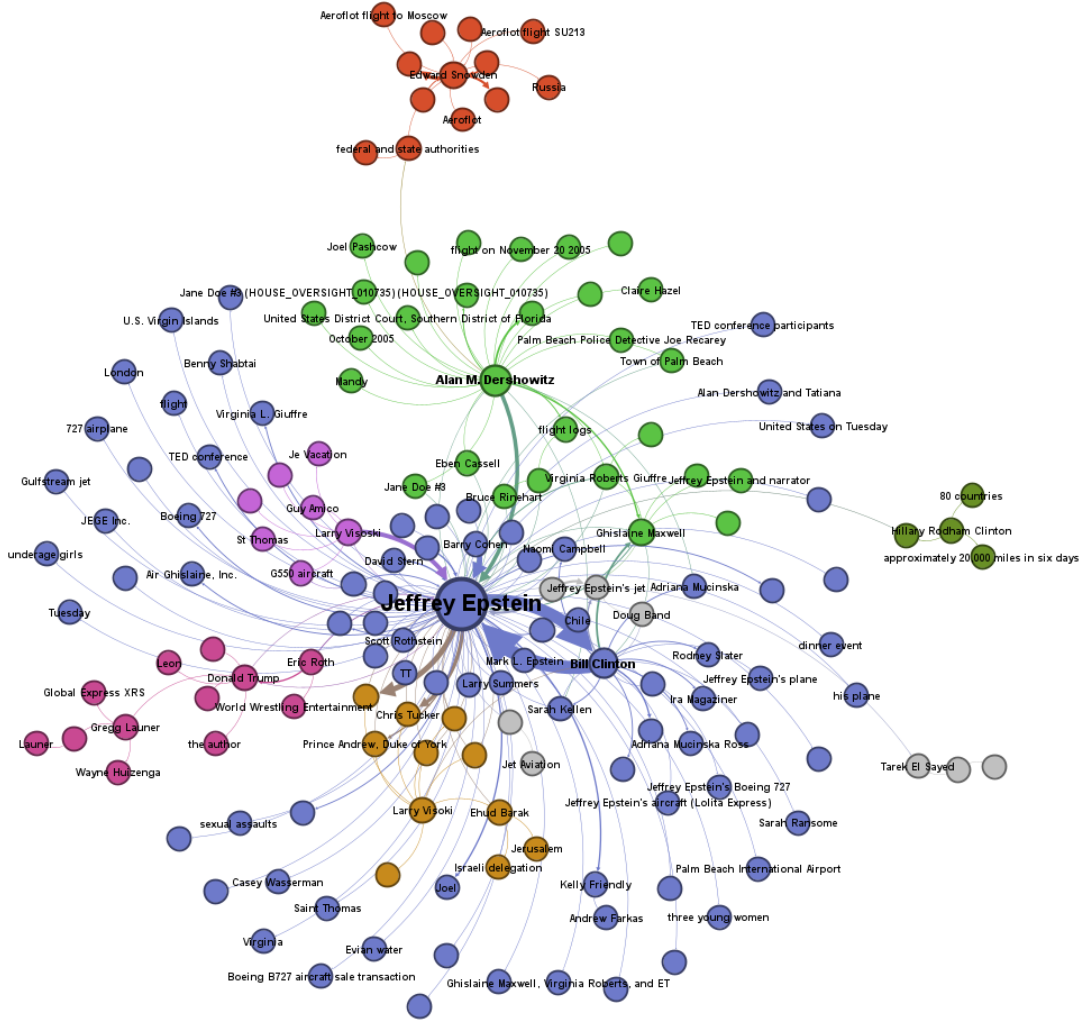


Figure 1: The complete aviation subgraph visualized using Gephi's ForceAtlas2 layout, with nodes colored by community membership. Jeffrey Epstein (center) is the gravitational hub of the entire network. Outliers are hidden.

The full network visualization reveals a strongly hub-and-spoke structure. Jeffrey Epstein dominates the center of the graph whose size visually dwarfs every other entity. The

distinct colored clusters radiate outward from this core, each representing a different facet of Epstein’s flight-related social world. Smaller peripheral clusters represent isolated subgraphs connected to the main network only through single bridge nodes.

3.3 Community Interpretations

Blue Community — The Core Epstein Aviation Network

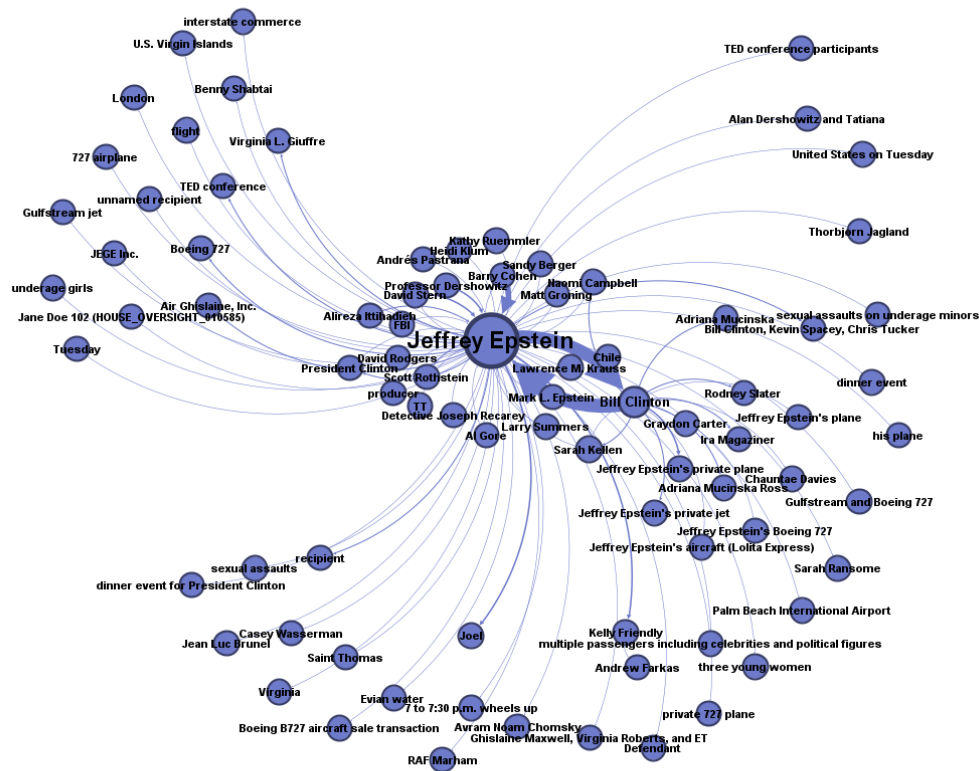


Figure 2: Blue Community — The Core Epstein Aviation Network

This is the largest and most central community, anchored by **Jeffrey Epstein** himself with a degree of 105. This community contains the primary cast of Epstein’s travel world: frequent passengers such as **Bill Clinton**, aviation assets like the **Boeing B727**, and companions such as **Adriana Mucinska** and **Sarah Kellen**. The blue cluster is not a tight social circle but rather Epstein’s personal hub, entities whose primary documented connection is to Epstein himself rather than to each other. Bill Clinton is the second-largest node within this community with a degree of 21, reflecting the extensive documentation of his flights on Epstein’s aircraft. The bidirectional edge between Clinton and Epstein (weights 27 and 25 respectively) is the single heaviest documented pair in the dataset, suggesting it was a focal point of legal inquiry.

Green Community — The Legal Cluster

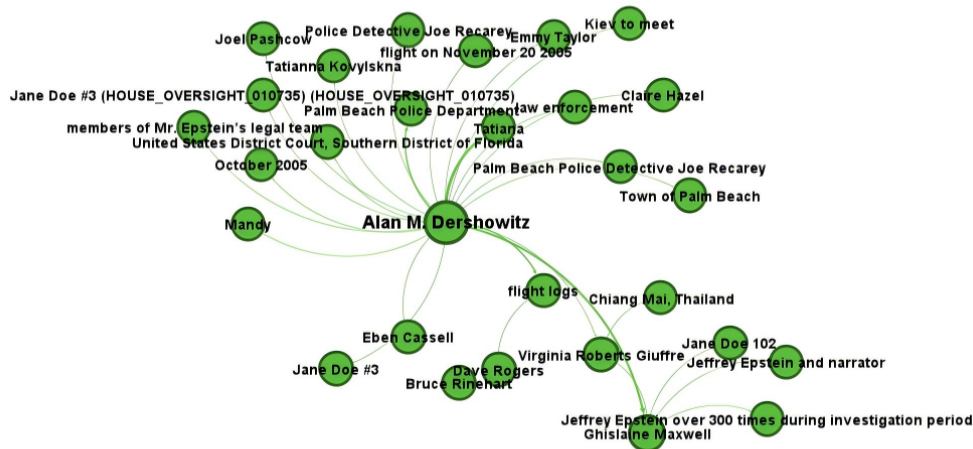


Figure 3: Green Community — The Legal Cluster

This 29-node cluster, centered on **Alan M. Dershowitz** as the prominent local hub, maps the network’s institutional legal defense infrastructure and judicial conflict zone. Dershowitz acts as a critical gatekeeper and defensive shield, bridging elite individuals with regulatory and investigative entities. The surrounding nodes reflect intensive legal friction and disclosure, with direct links to law enforcement and judicial bodies like **the United States District Court (Southern District of Florida)**. The connections to **Virginia Roberts Giuffre**, **flight logs**, and specific investigative markers (e.g., ‘**HOUSE_OVERSIGHT**’ documentation, **Jane Doe indices**) demonstrate that this sub-network was generated through aggressive civil discovery and legal cross-examinations. Ultimately, this green cluster highlights how institutional legal systems, police entities, and litigation documents intersect around a central legal broker to shape the visible landscape of the broader network.

Purple Community — The Pilot Operations Cluster

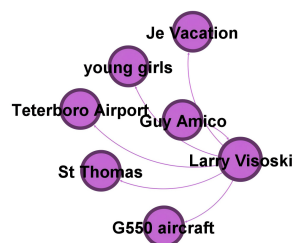


Figure 4: Purple Community — The Pilot Operations Cluster

This small but structurally significant 7-node cluster is centered on **Larry Visoski**, Epstein’s primary personal pilot for many years, who testified in multiple legal proceedings. His neighbors in this cluster include **G550 aircraft** (one of Epstein’s documented aircraft),

Teterboro Airport (a private aviation hub in New Jersey frequently used by Epstein), **St Thomas** (a Caribbean destination serving as the aerial transit checkpoint to Epstein’s private island), **Guy Amico** (another pilot), and **young girls** (victims). The direct edge from Visoski to young girls in the context of flights represents the trafficking purpose served by these aviation operations. Driven by personal pilots like Larry Visoski, the logistical pipeline funneled these young girls through St. Thomas as a strategic transit hub before transferring them via private helicopter or boat to Epstein’s secluded island, Little St. James.

Yellow Community — The VIP Passenger Cluster

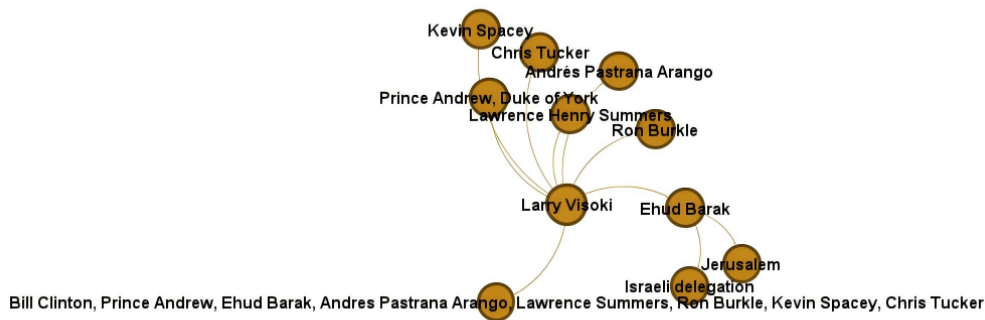


Figure 5: Yellow Community — The VIP Passenger Cluster

This 11-node community maps the network’s celebrity capital and geopolitical elite ties, centering again on **Larry Visoski** (a transcription variant in the source documents). The spokes radiating from Visoski include global political figures such as **Bill Clinton**, **Prince Andrew**, **Ehud Barak** (former Israeli Prime Minister), **Andrés Pastrana Arango** (former Colombian President), **Lawrence Henry Summers** (former U.S. Treasury Secretary), **Ron Burkle** (billionaire investor), **Kevin Spacey** (famous American actor), and **Chris Tucker** (Hollywood star). Visoski’s comprehensive piloting logs and deposition records captured how these otherwise disconnected global elites were structurally collapsed into a singular, tightly bound collective within the dataset due to their shared presence aboard the network’s private aviation fleet.

4 Graph Metrics

4.1 Metrics Overview

Gephi computed a range of centrality metrics for each node. Table 5 describes the main metrics and their interpretation in this network context.

Table 5: Centrality Metrics: Descriptions and Network Interpretations

Metric		Description	Network Interpretation
Degree		Total number of edges (in + out)	Overall connectivity; how many entities a node interacts with in flight-related documents
Betweenness Centrality		Number of shortest paths between all node pairs that pass through this node	Measures brokerage and bridge roles; high values indicate gatekeepers of information flow
Closeness Centrality		Inverse of average shortest path distance to all other nodes	How efficiently a node can reach or be reached by others
Eigenvector Centrality		Centrality weighted by the centrality of neighbors	Identifies nodes connected to other important nodes

4.2 Results Interpretation

Table 6 consolidates the key graph metrics for the 20 most-connected nodes (ranked by degree), allowing direct cross-metric comparison. Jeffrey Epstein obviously leads every metric by a wide margin because he is the center of this topic.

Table 6: Combined Centrality Metrics for Top 20 Nodes by Degree

Rank	Node	Degree	Betweenness	Closeness	Eigenvector
1	Jeffrey Epstein	105	12,057.51	0.610	1.000
2	Alan M. Dershowitz	27	4,452.57	0.455	0.267
3	Bill Clinton	21	1,970.23	0.414	0.206
4	Edward Snowden	11	1,605.00	0.255	0.024
5	Ghislaine Maxwell	10	634.25	0.436	0.188
6	Larry Visoki	9	241.50	0.304	0.104
7	Donald Trump	8	1,439.00	0.398	0.125
8	Larry Visoski	8	894.50	0.391	0.115
9	Virginia Roberts Giuffre	5	165.00	0.416	0.151
10	Ehud Barak	4	360.81	0.387	0.116
11	Gregg Launer	4	653.00	0.289	0.019
12	Sarah Kellen	4	103.08	0.426	0.152
13	Jeffrey Epstein's jet	3	5.07	0.295	0.039
14	Hillary Rodham Clinton	3	329.00	0.280	0.014
15	Ron Burkle	3	31.81	0.383	0.111
16	Kevin Spacey	3	31.81	0.383	0.111
17	Lawrence Summers	3	67.38	0.386	0.124
18	Eric Roth	3	165.00	0.391	0.115
19	Dave Rogers	3	58.33	0.382	0.116
20	Chris Tucker	3	31.81	0.383	0.111

Other key persons are interpreted as follows:

Alan M. Dershowitz is the second most structurally significant node despite not being a passenger in the traditional sense. His high betweenness (4,452) reflects his role bridging the personal flight world and the legal/investigative nexus, and his eigenvector centrality (0.267) is boosted by direct connection to Epstein.

Bill Clinton follows closely across all metrics, consistent with his status as the most thoroughly documented passenger. His high degree (21), high closeness (0.414), and eigenvector centrality (0.206) all confirm his centrality to the network as a politician. Federal Aviation Administration (FAA) flight logs explicitly substantiate his high connection density, documenting that the former President took at least 26 trips aboard Epstein’s private Boeing 727 (the “Lolita Express”) between 2001 and 2003.

Larry Visoski has a relatively high betweenness of 894.5, which captures his operational role as a logistical bottleneck. As the person who physically flew Epstein’s aircraft for years and who submitted to legal depositions about passengers, his records explicitly documented the pipeline of transferring young women and minors through the St. Thomas transit hub before forwarding them to Epstein’s private islands.

Ghislaine Maxwell, the chief accomplice to Epstein, functions as the network’s primary institutional bridge. Her betweenness (634.25) and closeness (0.436) reflect her dual roles: as daughter of the prominent British media tycoon, she did not only operate within the core, but also actively regulated the linkages between the illicit operational base and external political and financial elites. **Sarah Kellen**, the assistant and operations manager for Epstein, worked more as an insulated operational scheduler with high closeness of 0.426 but lower betweenness of 103.08.

Ron Burkle, **Kevin Spacey**, and **Chris Tucker** all share identical betweenness (31.81), closeness (0.383), and eigenvector centrality (0.111). This complete convergence indicates that they lack individual relational autonomy within the dataset but exist as a unified celebrity cluster whose connections are entirely synchronized. This is an empirical reflection of their shared 2002 African transit record aboard Epstein’s private jet.

Virginia Roberts Giuffre is characterized by low betweenness of 165 but high closeness of 0.416, a reflection of her position as a core victim within the network. Her direct, short-path exposure to key systemic hubs drags her to the center of the network, while she did not function as an operational gatekeeper or a strategic bridge coordinating connections between disparate clusters.

Donald Trump (betweenness 1,439, closeness 0.398, relatively low eigenvector centrality 0.125) exhibits a distinctly different position from Bill Clinton within the network architecture. Trump’s involvement is structurally constrained to the 1990s Palm Beach social circuit. His relationship with Epstein completely fractured in the mid-2000s following a 2004 Palm Beach real estate dispute, which prevented Trump from transitioning into late-stage hubs in the network.

Edward Snowden is a notable outlier with high degree (11) and betweenness (1,605) but very low eigenvector centrality (0.024) and closeness (0.255), acting as the most peripheral node in this table. His inclusion in the network is simply a procedural artifact of text co-mention, connected to the main network through only a single bridge node “federal and state authorities”, not relevant to the Epstein core.

5 Conclusion

This study applied a graph analysis to the aviation subnetwork of the Jeffrey Epstein document corpus, extracting 253 nodes and 246 unique edges connected through flight-related interactions. Using Gephi’s ForceAtlas2 layout for visualization and the Louvain algorithm for community detection, we identified several structurally meaningful communities including the Core Network, the Legal Cluster, the Pilot Operations Cluster, and the VIP Passenger Cluster.

The centrality analysis confirms that the network’s criminal infrastructure operates as a centralized, highly functional hierarchy mapped by five distinct roles around Jeffrey Epstein: the legal anchor around Alan M. Dershowitz, the social gatekeeper (Ghislaine Maxwell), the operators (Larry Visoski and Sarah Kellen), the global elites (Bill Clinton, Prince Andrew, Kevin Spacey, etc.) and the victims (Virginia Roberts Giuffre).

The private plane of Epstein was the connective tissue binding together politicians, pilots, attorneys, entertainers, and accusers into a single, structurally coherent social world defined, organized, and controlled by one man.